

UnityReach Microfinance Bank

Executive summary

This project covers an analysis on loan disbursed (**\$2.583B**) by UnityReach Microfinance Bank, to clients of different occupational backgrounds (Civil servants, Traders, Teachers, etc.), age, income, and payment term. A total of **\$818M** loan have been repaid during the period of this analysis, with an outstanding loan payment of **\$1.764B**.

The purpose of this analysis is to help the Head of Credit & Risk Management to understand their clients and also easily identify credit risk clients. This analysis would help the bank identify customers who are likely to default or delay repayment, thereby enabling it take proactive measures during loan processing, monitoring, and recovery.

Facts

- Almost equal balance of gender (**Male: 4,913, Female: 5,087**)
- Total loan disbursed: **\$2.583B**
- Total loan repaid: **\$818M**
- Unpaid balance: **\$1.764B**
- Default rate: **63.74% (Male: 64.42%, Female: 63.08%)**

Task 1: Analyze past customer loan behavior

♣ Income Analysis

Facts

Income earners (\$170,000 - \$240,000) have the lowest default rate (45.42%)

Income earners (\$30,000 - \$100,000) have the highest default rate (92.58%)

Highest client base (\$100,000 - \$170,000) – **2,222 clients**

Lowest client base (\$310,000 - \$380,000) – **1,222 clients**

Findings

Generally, there is an almost equal distribution of clients between earners from \$30,000 to \$310,000 (over 2,000 clients). Only clients earning above \$310,000 had the fewest number of clients (1,222) and the lowest loan portfolio.

Insights

Loan Disbursed	Loan repaid \$ (%)	Client repayment (%)	Income Range
559M	2%	7%	30,000-100,000
579M	15%	20%	100,000-170,000
556M	49%	54%	170,000-240,000
567M	49%	53%	240,000-310,000
321M	52%	55%	310,000-380,000

People who earn between **\$310,000** and **\$380,000** generally have the lowest loan portfolio of **\$ 321 million** and have paid back **\$ 168 million** (i.e., **52%** repayment). Of the **1,222** clients in this category, **667** have repaid, i.e., **55%** of the clients in the group have repaid their loans. Clients in this category have the best client repayment rate and the best loan repayment rate. This suggests likely that the higher the client's income, the better their probability of loan repayment (other factors kept constant).

Clients (Income bracket \$30,000 - \$100,000)

	Total Clients	Clients paid	%Repayment by clients
Civil Servant	416.0	38.0	9.13%
Teacher	469.0	40.0	8.53%
Trader	441.0	32.0	7.26%
Entrepreneur	415.0	26.0	6.27%
Student	443.0	26.0	5.87%

Clients within the income bracket of **\$30,000 to \$100,000** have the worst repayment rate, as can be easily seen from the image above. Each occupation has a repayment percentage of less than 10%, which is extremely alarming.

Recommendations

- **Portfolio Optimization by Income Segment:** Reduce loan portfolio size of clients within the \$30,000 - \$100,000 category, as their loan repayment rate and client repayment rate are very low, and prioritize high earners
- **Selective Lending Within the Segment:** Where loans are considered for clients earning between \$30,000 and \$100,000, priority should be given to civil servants and teachers with adequate collateral coverage, as these profiles present comparatively lower credit risk than other occupations

♣ Occupation Analysis

Findings

Civil servants have the best client repayment rate (**37.59%**) and the lowest default rate (**62.41%**, even though this is still high).

Traders have a client repayment rate (**36.89%**) and a default rate of (**63.11%**)

Entrepreneurs, students, and teachers all had the highest default rate (**above 64%**).

Insights

Despite Entrepreneurs, students, and teachers having the highest default rate (**above 64%**), they also have the lowest repayment rate (**below 36%**).

% Composition of Clients by salary range (Student, Teachers, Entrepreneur)

Occupation	\$30,000-\$100,000	\$100,000-\$170,000	\$170,000-\$240,000	\$240,000-\$310,000	\$310,000-\$390,000
Student	22%	23%	21%	21%	14%
Teacher	23%	22%	22%	22%	11%
entrepreneur	21%	23%	22%	22%	12%

Repayment percentage for each salary range (Student, Teachers, Entrepreneur)

Occupation	\$30,000-\$100,000	\$100,000-\$170,000	\$170,000-\$240,000	\$240,000-\$310,000	\$310,000-\$390,000
Student	6%	18%	59%	52%	52%
Teacher	7%	22%	54%	50%	54%
entrepreneur	6%	20%	50%	58%	50%

A look at the tables above suggests that, despite having slightly more clients within the income range of \$30,000 - \$100,000 (across students, teachers, and entrepreneurs), the repayment percentage within this group is very poor. In contrast, individuals in these occupations who earn above \$170,000 have a repayment percentage of 50% or higher.

Recommendations

- **Loan Prioritization by Occupation:** While civil servants and traders may be prioritized in loan disbursements, it is essential to ensure sufficient collateral coverage before approval, as a default rate exceeding 60% indicates elevated credit risk.
- **Income-Based Assessment for Other Groups:** For entrepreneurs, teachers, and students, loan approvals should emphasize clients with higher income levels, as they demonstrate better repayment capacity.

♣ Loan size

Facts

Influence of income on loan repayment

	Total Clients	Clients paid	%Repayment by clients
30000 - 99999	2,184	162	7.42%
100000 - 169999	2,222	454	20.43%
170000 - 239999	2,188	1,181	53.98%
240000 - 309999	2,184	1,162	53.21%
310000 - 380000	1,222	667	54.58%

The image above shows the influence of income on loan repayment. People who earn lower tend to have lower repayment rates in comparison to clients of higher income.

Findings

A quick view into the loan portfolio between \$420,000 to \$500,000 per each income group.

	Total Clients	Clients paid	%Repayment by clients
30000 - 99999	334.0	0.0	0.00%
100000 - 169999	359.0	52.0	14.48%
170000 - 239999	344.0	166.0	48.26%
240000 - 309999	391.0	186.0	47.57%
310000 - 380000	209.0	107.0	51.20%

There were more clients (**334**) who earn less than **\$100,000** than clients (**209**) who earn above **\$310,000** who took loan above \$420,000. The repayment rate of these clients – who earn less than \$100,000, is basically 0 i.e., there is a high- risk alert.

This was also observed in the case where these clients took loan above \$100,000, the client repayment rate was 0.

Insights

For clients with lower income, taking a larger loan portfolio implies a longer repayment term – the longer the repayment term, the higher the chances of default.

Recommendations

- **Staggered Lending for Low-Income Clients:** For low-income clients requesting large loan amounts, implement phased disbursement and structured repayment plans by splitting the portfolio into smaller tranches, with subsequent loans granted only after full repayment of the previous tranche.

- **Income-Based Risk Consideration for Large Loans:** For large loan portfolios, prioritize higher-income clients over lower-income clients, as higher-income earners demonstrate stronger repayment performance and reduced default risk